

Estimation Process

The pre-trade of pre-trade cost estimation process is described through the following example. For illustrative purposes, we use two stocks and three brokers. In practice, investors are encouraged to use a much larger sample of stocks with various order sizes and transaction strategies, and solicit cost estimates from more than three brokers.

Analysts can also implement this approach without revealing their hand to their brokers simply by using a financial data provider such as Bloomberg since many brokers have embedded their pre-trade models into these financial systems. Managers can generate a large enough sample of trades through these systems to calibrate the pre-trade of pre-trade model without their brokers becoming any the wiser or learning their true intentions (also see Kyle, 1985).

This approach will also allow analysts to test and critique broker models from various perspectives. Our pre-trade of pre-trade process is as follows:

Step I: Select Preferred Market Impact Model

The first step in the process is to select the preferred market impact model. In this example, we use a simplified version of the I-Star model that does not separate temporary and permanent impact. Analysts are encouraged to experiment with different cost models to determine the formulation that works best for their needs. This model is:

$$MI_{bp} = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3} \cdot POV^{a_4} \quad (11.1)$$

Step II: Solicit Broker-Dealer Cost Estimates

The second step in the process is to collect cost estimates for a universe of stocks from different brokers. Be sure to use a large enough range of stock characteristics, sizes, and trading strategies.

It is important to note that the vendors who are providing the cost estimates for the estimation process are likely to have constructed their models from their own trade data. In these cases, the model may be well suited their client's trading styles but it may not be as accurate for different trading styles. There is the potential that this model would have a very good in-sample fit but may not be as accurate out-of-sample (suffer from in-sample bias). An example of this is as follows. Suppose broker A's clients consist only of traditional long-term buy and hold indexer managers who usually incur low impact cost, and broker B's clients consist only of short-term trading funds who typically transact in a high cost